The Quarter

An Unrepresentable Number, and Why It Is a Choice

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The fourth card: the datum the other three compare.

“Can you find the polynomial in the physical model of this device?”

One choice from the knower; the question the reader has to solve.

Reading note

This volume studies finite record systems that carry only a projection $\pi_R(q)$ of a datum q, and the distinction between an operational computation, which a record performs, and a selection from the fiber of π_R, which it does not. The object of the volume is such a datum, the *quarter* (Definition 1.4). It is defined relative to a specified family of record readers, by the relations every record of the family bears to it: a formula that the family's own readers evaluate would be a record of the family and would represent it.

Section 1 fixes records, projections and receipts, and separates three claims that the rest of the volume keeps apart (Remark 1.5). Section 2 works three projection-and-selection problems in full: the cosmological constant in unimodular gravity, the omitted term of Jacobi's theta function, and the constant of Kolmogorov invariance. Section 3 proves that a value of the quarter can be supplied only by a selection from a fiber. Section 4 records two bounds: one on the length of the record that reaches an integer, one on what a fixed formal system certifies about minimal descriptions. Section 5 runs the argument in the opposite direction: under comparability, the Riemann hypothesis and the separation of P from NP become countable families of comparisons. Section 6 formalizes the instances in which the quarter appears as an omitted term or a floor, and Section 7 locates it in each of the three cards.

Definition 0.1 (Calibration)**.** A *calibration* of a statement is the stated source of its warrant, reported with the statement as part of its reading. This volume uses five.

Standard
The statement is cited from the literature with its hypotheses.

Recorded
The statement is the conclusion of a deductive sequence written here, each premise of which is a definition of this volume or a cited result.

Proved here
The statement is proved here from definitions alone, with no cited premise.

Measured
The statement is extracted from the declarations of the middle card, or observed in a run of it; the text says which. It is not a laboratory measurement unless a citation says so.

Proposed
The statement is an identification or continuation supplied by the reader and not derived. A proposed statement asserts that an isomorphism exists, between the numbering of the construction's descriptions by the natural numbers and the description of what it is identified with, that the machine can exploit as a computation. Both sides are finite records: the reading is a computable number, and the reference is a value with its uncertainty or a stated structure. The statement is therefore one the machine can evaluate, a case that could be false, and the comparison that would revise it is that evaluation. What no evaluation supplies is that the reference's finite record stands for what it records, as a finite halt stands for its limit.

A recorded statement is a record in the sense of Definition 1.1: a finite sequence of steps. That the record is mathematics, true and about its object, is the reader's reading of it.

The companion volumes are the geometric card, which fixes what must be retained, and the numerical card, which fixes what may be claimed from a reading; the middle card is a construction checked by a proof assistant, and its two record chapters check that the record carries both. No declaration of the middle card appears here by name; where its declarations are the evidence, the statement is calibrated *measured*. The obstruction is representational, not ontological, and relative to a reader: the root of the numerical card is finitely specified as an algebraic number and has no terminating decimal, the fourth coordinate of the geometric card is a real number with no image under the observation, and the name of the middle card is a symbol with no derivation.

1 Records, projections, receipts

Throughout, Σ is a finite alphabet, Σ^* the free monoid of finite words over Σ with empty word ϵ, $u \cdot v$ the concatenation of words, and $|u|$ the length of u.

Definition 1.1 (Record; record system)**.** A *record system* is a triple $\mathcal{R} = (\Sigma, \to, \mathrm{ev})$ consisting of a finite alphabet, a step relation $\to \subseteq \Sigma^* \times \Sigma^*$ with $|w'| = |w| + 1$ whenever $w \to w'$, and a partial map $\mathrm{ev} : \Sigma^* \rightharpoonup V$, the *reader*, into a value set V. A *record* of $\mathcal{R}$ is a word reached from ϵ by finitely many steps. The *length* of a record is the number of steps taken to write it, and equals its word length.

The step relation is an operational transition system on Σ^*: each step writes exactly one symbol. The reader is the only means by which a record denotes a value.

Definition 1.2 (Representable)**.** Let $Q \subseteq V$. A datum $q \in Q$ is *representable* in $\mathcal{R}$ if some record w of $\mathcal{R}$ determines it, that is, $\mathrm{ev}(w) = q$.

Definition 1.3 (Projection; receipt)**.** Let Q be a domain and $\pi_R : Q \to Y_R$ an *observation map* into the values written by the records of $\mathcal{R}$. The record system *carries a projection of* $q \in Q$ if $\pi_R(q)$ is written in some record of $\mathcal{R}$ and the fiber $\pi_R^{-1}(\pi_R(q))$ contains an element other than q; in particular π_R is not injective. A *receipt* on a declared domain $D \subseteq Q$ is a map $\rho : D \to \mathcal{A}$ into an auxiliary alphabet such that

$$(\pi_R, \rho) : D \to Y_R \times \mathcal{A}$$

is injective. An *exact reader* is a left inverse of (π_R, ρ) on its image.

With the encoding taken to be the identity, this is the receipt of the geometric card (its Definition "Receipt; exact reader"): an exact reader exists if and only if ρ is a receipt, and it is then unique on the image. A receipt is an argument of the inverse map; the observation alone determines only the fiber.

Definition 1.4 (The quarter)**.** Let $\mathcal{F} = \{\mathcal{R}_\alpha\}_{\alpha \in I}$ be a family of record systems, with observation maps $\pi_\alpha : Q \to Y_\alpha$, closed under finite concatenation: if w is a record of $\mathcal{R}_\alpha$ and w' a record of $\mathcal{R}_\beta$, then $w \cdot w'$ is a record of some member of $\mathcal{F}$. A datum $q \in Q$ is a *quarter* for $\mathcal{F}$ if

(i) *universal projection:* every $\mathcal{R}_\alpha$ carries a projection of q under π_α;

(ii) *non-representability:* q is representable in no $\mathcal{R}_\alpha$, so that no finite record of the family determines q by evaluation;

(iii) *exterior selection:* a value of q is supplied, when it is supplied, by a choice of an element of a fiber $\pi_\alpha^{-1}(\pi_\alpha(q))$.

The definition states the relations every record bears to q and contains no formula for q. Theorem 3.1 shows that this is forced: clause (iii) follows from clauses (i) and (ii) and the closure hypothesis. It is retained in the definition to name the act.

Remark 1.5 (Three claims). For a datum q three claims are kept distinct:

(1) a given observation π_R does not determine q: its fiber through q contains another element;

(2) a specified family $\mathcal{F}$ of record readers cannot represent q: clause (ii) of Definition 1.4;

(3) no finite description, in any language, specifies q.

Then (1)$\not\Rightarrow$(2)$\not\Rightarrow$(3). The traceless field equations of Section 2.1 do not determine Λ, yet the record "$\Lambda = 0$" writes one value of it. The root $\alpha = (18+3\sqrt{10})/13$ of the numerical card has no terminating decimal, so no reader of decimal truncations represents it, yet the expression just written is a finite description that specifies it exactly. Clause (i) of Definition 1.4 is a claim of form (1), clause (ii) of form (2); this volume makes no claim of form (3). *Proved here.*

2 Three projections

Each of the three constructions below is presented as a projection-and-selection problem: a domain Q, an observation that the record carries, the fiber the observation leaves, and the datum that selects a point of it. The three obstructions differ, and Section 2.4 compares them.

2.1 The integration constant

Let g be a Lorentzian metric on a connected four-dimensional region, and restrict to charts in which $\sqrt{-g} = 1$. Varying the connection freely and the metric only along traceless directions, the stationary equations of the first-order vacuum action are

$$\Gamma = \Gamma(g), \qquad R_{\mu\nu} - \tfrac{1}{4} g_{\mu\nu} R = 0.$$

The contracted Bianchi identity $\nabla^\mu(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = 0$, applied to the traceless equations, gives $\partial_\nu R = 0$. Hence $R = 4\Lambda$ for a constant Λ, and the field equations are

$$R_{\mu\nu} = \Lambda\, g_{\mu\nu}.$$

The constant enters as a constant of integration, supplied as boundary data, and not as a coefficient of the action: this is the unimodular presentation of gravity [9], and the derivation is carried out in the numerical card beside its gravity model. *Standard.*

As a fiber projection: the domain is the set of admissible values $\Lambda \in \mathbb{R}$; the record is the action together with its traceless stationary equations; the observation forgets Λ, and its fiber is all of $\mathbb{R}$. The solution carries $\Lambda g_{\mu\nu}$, the image of the constant in the field. The receipt is the datum that fixes it: boundary data or an observation outside the action. The equations determine that Λ, once fixed, is carried unchanged; they do not determine its value. This is claim (1) of Remark 1.5 for the field equations.

2.2 The omitted zero

Riemann's function is $\psi(x) = \sum_{n\geq 1} e^{-n^2\pi x}$ for $x > 0$ [16]. Jacobi's transformation of the theta function $\theta(x) = 1 + 2\psi(x) = \sum_{n\in\mathbb{Z}} e^{-n^2\pi x}$ is $\theta(1/x) = \sqrt{x}\,\theta(x)$, and in terms of ψ it reads

$$\psi(1/x) = \sqrt{x}\,\psi(x) + \frac{\sqrt{x}-1}{2}. \tag{1}$$

The term $(\sqrt{x}-1)/2$ is the boundary defect between the sum over $n \geq 1$ and the sum over $n \in \mathbb{Z}$: it is the image of the $n = 0$ term that ψ omits and θ keeps. *Standard* [6, Ch. 1].

Proposition 2.1 (The omitted term is a residue)**.** *For* $\operatorname{Re} s > 1$,

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty \psi(x)\, x^{s/2-1}\,\mathrm{d}x,$$

and splitting the integral at $x = 1$ *and applying* (1) *on* $(0, 1)$ *gives*

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \psi(x)\left(x^{s/2-1} + x^{(1-s)/2-1}\right)\mathrm{d}x.$$

The integral is entire and invariant under $s \mapsto 1-s$*. The poles of the left side at* $s = 0$ *and* $s = 1$*, and in particular the pole of* ζ *at* $s = 1$ *with residue* 1*, come entirely from the omitted term.* Standard; the computation is reproduced here.

Proof. Termwise, $\int_0^\infty e^{-n^2\pi x}x^{s/2-1}\,\mathrm{d}x = \pi^{-s/2}\Gamma(s/2)n^{-s}$, and the sum converges absolutely for $\operatorname{Re} s > 1$. On $(0, 1)$ substitute $x \mapsto 1/x$ and use (1):

$$\int_0^1 \psi(x)x^{s/2-1}\,\mathrm{d}x = \int_1^\infty \psi(x)x^{(1-s)/2-1}\,\mathrm{d}x + \frac{1}{2}\int_1^\infty \left(x^{-s/2-1/2} - x^{-s/2-1}\right)\mathrm{d}x.$$

The last integral equals $\frac{1}{s-1} - \frac{1}{s} = \frac{1}{s(s-1)}$ for $\operatorname{Re} s > 1$. The remaining integrals converge for all s since $\psi(x) = O(e^{-\pi x})$, and their sum is symmetric under $s \mapsto 1-s$ by inspection. □

As a fiber projection: the record is the count beginning at $n = 1$; the observation forgets the $n = 0$ term; the projection that ζ carries of it is the residue 1 at $s = 1$. The receipt is the choice of the origin of the count, and the omitted term is explicitly computable: it is the constant $\frac{1}{2}$ of Proposition 2.4(b). The self-dual point $x = 1$ of (1) is the point at which the two halves of the split integral are exchanged, and it is the only point at which the defect enters the computation.

2.3 The simulation constant

Let U and V be universal machines and $K_U(x)$, $K_V(x)$ the lengths of the shortest programs producing x on each.

Proposition 2.2 (Invariance)**.** *There is a constant* c_{UV}*, depending on* U *and* V *and not on* x*, with* $K_U(x) \leq K_V(x) + c_{UV}$ *for all* x*.* Standard *[10, 12].*

Proof. c_{UV} is the length of a program for U that simulates V; prefix it to any V-program for x. □

Proposition 2.3 (No system certifies the constant)**.** *For every consistent formal system S with a computable set of axioms there is a constant c_S such that S proves no statement of the form $K_U(x) > c_S$, although such statements are true for all but finitely many x.* Standard *[2].*

As a fiber projection: the domain is the set of universal reference machines; every length written on a universal machine is the image of the description length up to an additive constant; the fiber is the set of constants compatible with Proposition 2.2. The choice of the pair of machines, and of an interpreter, supplies an admissible c_{UV}: the interpreter's length, written in a finite record. The constant c_S of Proposition 2.3 is a different object, a constant of a formal system bounding which complexity lower bounds that system proves. Proposition 2.3 does not show that no record certifies a particular interpreter's overhead, and the two constants are not identified here.

2.4 One object

Proposition 2.4 (Affine freedom of the three constants)**.** *Each of the three data above enters its record additively, and the record's internal identity is satisfied, in its affine form, for every value.*

(a) The action and the traceless stationary equations of Section 2.1 do not contain Λ; the contracted Bianchi identity forces R to be constant and leaves its value free, and every real Λ is attained by a solution.

(b) For every $c \in \mathbb{R}$ the function $\psi_c = \psi + c$ satisfies

$$\psi_c(1/x) = \sqrt{x}\,\psi_c(x) + (\tfrac{1}{2} - c)(\sqrt{x} - 1), \qquad x > 0.$$

The defect vanishes exactly at $c = \frac{1}{2}$, where $\psi_{1/2} = \theta/2$ is the count that begins at zero.

(c) For every pair of universal machines the invariance inequality holds with every constant $c \geq c_{UV}$; its proof exhibits one admissible constant, an interpreter's length, and the inequality singles out none.

In (a) and (c) no internal identity singles out a value. In (b) the homogeneous form of Jacobi's law singles out exactly the omitted term, and the record ψ, which counts from one, does not carry it. Recorded*, from the three preceding cited results [9, 6, 10, 12].*

Proof. (a) is the derivation of Section 2.1; de Sitter, Minkowski and anti-de Sitter space realize positive, zero and negative Λ. (b) By (1), $\psi_c(1/x) - \sqrt{x}\,\psi_c(x) = \frac{1}{2}(\sqrt{x}-1) + c - c\sqrt{x} = (\frac{1}{2} - c)(\sqrt{x}-1)$, which vanishes identically in x if and only if $c = \frac{1}{2}$; and $\psi + \frac{1}{2} = \theta/2$. (c) If $K_U \leq K_V + c_{UV}$ then $K_U \leq K_V + c$ for every $c \geq c_{UV}$. □

The three are a family of projection-and-selection problems with different obstructions. In (a) the value is fixed by data outside the action; in (b) the omitted term is explicitly computable, $c = \frac{1}{2}$; in (c) the constant is fixed by choosing the machines and an interpreter. *Proposed*: that the three constants are one invariant, the quarter, read by three readers, each the image of a single act, the choice, under a different observation. The identification is not numerical equality, which has no meaning across three units. It requires a correspondence among the three fibers that preserves more than the shared vocabulary of projection and receipt; that correspondence is the missing object, and it is not constructed here.

3 The quarter is a choice

Theorem 3.1 (Unrepresentable means chosen)**.** *Let q satisfy clauses (i) and (ii) of Definition 1.4 for a family $\mathcal{F}$ of record systems closed under finite concatenation. Then no procedure expressible in $\mathcal{F}$ supplies a value of q. Any act that supplies one selects an element of a fiber $\pi_\alpha^{-1}(\pi_\alpha(q))$ by no procedure of the family: it is a choice from the fiber.* Recorded*, from Definition 1.4 and the closure hypothesis; the step is the standard one, that a selection procedure inside the system would make the value representable there [2, 12].*

Proof. Suppose a procedure P expressible in $\mathcal{F}$ supplied a value of q. The description of P followed by its finite run is, by closure under finite concatenation, a record of a member of $\mathcal{F}$, and the reader of that member returns q. This contradicts clause (ii). Hence no such P exists. The set of candidate values is definite: by clause (i) every $\mathcal{R}_\alpha$ writes $\pi_\alpha(q)$, and the fiber over it is a determined subset of Q. A selection of an element of a determined set by no procedure of the family is what this volume calls a choice. □

Definition 1.4 states the quarter by its relations because a definition is a record. The middle card's proof assistant has three relevant axioms: propositional extensionality, a quotient axiom from which function extensionality follows, and a choice axiom. The middle card's receipts bill the first two. No step of its reading bills the choice axiom, and the one declaration that reaches it does so only through an automatic arithmetic procedure, not through any selection the card makes. *Measured*, extracted from a sweep of its declarations.

Theorem 3.2 (Diaconescu)**.** *Over intuitionistic logic, the axiom of choice together with propositional extensionality and function extensionality implies the law of excluded middle.* Standard *[5].*

The middle card therefore keeps two of the three hypotheses of Theorem 3.2 and does not use the third. Every question it answers, it answers by a decision procedure it carries, and not by excluded middle. By Theorem 3.1 the quarter is supplied only by a selection; the construction executes none. The selection of Theorem 3.1 is of one element from one nonempty fiber, and the axiom of choice, a logical principle about families of sets, is not needed for it. *Proposed*: that the quarter is exactly the choice step the construction does not execute, the explicit selection read as the axiom the card does not bill.

4 Two bounds

The upper bound states that every integer is reached by a finite record of controlled length. The lower bound states that the certification of a minimal description past a fixed constant is not reached by any proof in a fixed system.

4.1 Upper bound: unique factorization

Theorem 4.1 (Fundamental theorem of arithmetic)**.** *Every integer $n \geq 2$ is a product of primes, uniquely up to order.* Standard *[7, Thm. 2].*

Proposition 4.2 (Representable integers have short records)**.** *Let $n = p_1^{e_1} \cdots p_k^{e_k}$ with distinct primes p_i and exponents $e_i \geq 1$, and write $\Omega(n) = \sum_i e_i$. Then $\Omega(n) \leq \log_2 n$, and the factorization*

record of n, the list of pairs (p_i, e_i) written in binary, contains at most $3\log_2 n$ binary digits. Hence every integer $n \geq 2$ is determined by a record whose length is linear in the number of binary digits of n. Recorded*, from the fundamental theorem of arithmetic [7, Thm. 2].*

Proof. Each prime is at least 2, so $2^{\Omega(n)} \leq n$. The number of binary digits of p_i is $\lfloor \log_2 p_i \rfloor + 1 \leq 2\log_2 p_i$, since $p_i \geq 2$; summed over i, and using $e_i \geq 1$, this is at most $2\sum_i e_i \log_2 p_i = 2\log_2 n$. The number of binary digits of e_i is $\lfloor \log_2 e_i \rfloor + 1 \leq e_i$; summed, this is at most $\Omega(n) \leq \log_2 n$. The total is at most $3\log_2 n$. Uniqueness in Theorem 4.1 makes the record a function of n, and multiplication recovers n from it. □

An integer is thus reached by a finite product, and its record is a finite polynomial in its digits. The quarter, relative to its family $\mathcal{F}$, is not among such data: by clause (ii) no record of $\mathcal{F}$ returns it. Where the family's readers are truncations of a series $\sum_k a_k x^k$, as decimal readers are, a datum with no terminating expansion is read by every truncation and returned by none, although, like the root α, it may have a finite specification in another language (Remark 1.5). *Recorded*, with the identification of the quarter as the limit of such a series *proposed*.

4.2 Lower bound: the cost of a minimal bound

The separation question asks whether every problem whose solutions can be checked in polynomial time can also be solved in polynomial time [3]. A proof of separation must exhibit, for some checkable problem, a lower bound on the size of every solver.

Proposition 4.3 (A certified minimal bound is uncertifiable)**.** *Let S be a consistent formal system with computable axioms and c_S its constant from Proposition 2.3. There is no proof in S that proceeds by certifying, for infinitely many inputs x, that no description of a solver's behaviour on x is shorter than a stated bound exceeding c_S.* Recorded*, from Proposition 2.3 [2].*

Proof. Such a certificate for x is a proof in S of $K_U(x') > c_S$ for the string x' encoding the solver's behaviour on x, which Proposition 2.3 excludes. □

A proof that proceeds by exhibiting a minimal representation and certifying its minimality is therefore blocked at a constant that the proving system cannot certify past. The lower bound has the shape of the upper one with the inequality reversed: the certification of a minimal bound is a countable series, no finite truncation of which certifies it, and the number at which the series would close is c_S, which is not certified in S.

Remark 4.4 (A distinct barrier)*.* The separation question also faces the natural-proofs barrier: if pseudorandom function generators of sufficient hardness exist, no property of Boolean functions that is constructive and large is useful against polynomial-size circuits [15]. That barrier concerns natural properties and pseudorandom functions; Proposition 4.3 concerns the incompleteness constant of a formal system. The two are distinct, and neither is derived from the other here. *Standard.*

Remark 4.5 (Scope of the lower bound)*.* Proposition 4.3 decides the separation question in neither direction. A bound that cannot be certified is not a bound that fails, and a constant that is not represented is not absent. The proposition states that the question is not settled by a count, and that every record of an attempted answer carries the projection of a choice it does not make. *Recorded* for the negative statement. A reading of it as an answer to the separation question is not proposed in this volume.

5 Comparability

Sections 3 and 4 proceed from the quarter's non-representability in its family to the shape of two bounds. This section proceeds in the opposite direction. Call a real datum q *comparable* if, for each rational number, the question whether it lies below or above q is answered, and call the set

$$O_q = \{\langle a, b\rangle : a \in \mathbb{Z},\ b \geq 1,\ a/b < q\},$$

with $\langle a, b\rangle$ a fixed binary encoding of the pair, the *comparison oracle* of q. Comparability is weaker than representability, since an oracle for the order need not produce the number. The section states the cited equivalences, then the decision problems they concern, then the proposed reduction.

The cited equivalences.

Theorem 5.1 (The Riemann hypothesis as a countable family of comparisons)**.** *Let $\sigma(n)$ be the sum of the divisors of n and γ Euler's constant. The Riemann hypothesis holds if and only if*

$$\sigma(n) < e^{\gamma}\, n \log\log n \qquad \text{for every } n > 5040.$$

Equivalently [11], with $H_n = \sum_{k \leq n} 1/k$, if and only if $\sigma(n) \leq H_n + e^{H_n} \log H_n$ for every $n \geq 1$, with equality only at $n = 1$. Standard *[17, 11].*

Theorem 5.2 (Positivity form)**.** *Let $\lambda_n = \sum_{\rho}\big[1 - (1 - 1/\rho)^n\big]$, the sum over the nontrivial zeros of ζ taken in conjugate pairs. The Riemann hypothesis holds if and only if $\lambda_n \geq 0$ for every $n \geq 1$.* Standard *[13].*

Theorem 5.1 is a countable family of comparisons against a constant, e^{γ}, that is given as a limit, $\gamma = \lim_{n\to\infty}(H_n - \log n)$; each comparison is made against the limit and not against a truncation of it. Theorem 5.2 is the order-dual form: a countable family of sign conditions. The functional equation of Proposition 2.1 is invariant under $s \mapsto 1 - s$, the symmetry induced by $x \mapsto 1/x$, whose fixed line is $\operatorname{Re} s = \frac{1}{2}$; the hypothesis places every zero of the entire part on that line, which is the same symmetry read as an order. *Recorded*, as a reading of the cited theorems.

Theorem 5.3 (Baker–Gill–Solovay)**.** *There is an oracle A with $\mathrm{P}^A = \mathrm{NP}^A$ and an oracle B with $\mathrm{P}^B \neq \mathrm{NP}^B$.* Standard *[1].*

The comparison oracle O_q is an oracle in the sense of Theorem 5.3, and the theorem states that the answer to the separation question depends on the oracle supplied, and that some oracle collapses it. *Recorded*, from Theorem 5.3 [1].

The decision problems.

Proposition 5.4 (Quantifier shape of the two questions)**.** *The Riemann hypothesis is equivalent to a Π_1^0 sentence: for every n, a decidable comparison of the form of Theorem 5.1 holds. The statement $\mathrm{P} = \mathrm{NP}$ is a Σ_2^0 sentence: there is a polynomially clocked machine M such that for every input x, M decides satisfiability correctly on x. Its negation $\mathrm{P} \neq \mathrm{NP}$ is Π_2^0: for every polynomially clocked machine M there is an input x on which M fails, a $\forall\exists$ sentence with decidable matrix. Once a witness M is supplied, the remaining sentence $\forall x\, \varphi(M, x)$ is Π_1^0, the shape of the Riemann hypothesis: a chosen object is a given one.* Recorded, *as a statement of quantifier shape; the classification of the Riemann hypothesis as a Π_1^0 sentence is* standard *[11].*

The proposed reduction. *Proposed*: that the Riemann hypothesis is the assertion that the omitted-zero projection of the quarter is comparable, every time, to the same side. *Proposed*: that the comparison oracle of the quarter is a collapsing oracle, $\mathrm{P}^{O_q} = \mathrm{NP}^{O_q}$ with q the quarter's real projection.

Remark (The proposed equivalence). *Proposed*: the two questions are one question read from two records, namely the Riemann hypothesis holds if and only if $\mathrm{P}^{O_q} = \mathrm{NP}^{O_q}$ for the quarter's real projection q, each being the statement that the quarter is comparable. The recorded content is weaker. Both questions are countable families of comparisons; the first asks whether the comparisons all fall one way, the second whether a supplied comparison oracle collapses the separation. The choice that would supply the quarter is the same act in both, and neither record performs it. The proposal would be discharged by a reader taking each comparison of Theorem 5.1 to a step of a bounded trace and back, with the oracle of Theorem 5.3 as O_q in both. The recorded statement common to both is that neither question is settled by a count, and that each is settled, if at all, by a comparison with the number the count does not write.

5.1 The last question

Every question the middle card answers comes with a receipt that a checker reads in time proportional to its length: a search writes the ledger down, and a check confirms it line by line. That is a certificate and a polynomial verifier. *Measured*, extracted from the card's construction. The question is what lies past every such receipt.

Proposition 5.5 (A new question has no polynomial verifier)**.** *Suppose a record system answers satisfiability and is closed under polynomial-time reduction. Then any yes-or-no question it cannot already answer is either decidable and outside NP, or undecidable. In either case it has no polynomial verifier.* Recorded*, from the Cook–Levin theorem [3] and the definition of NP.*

Proof. If the question were in NP it would reduce in polynomial time to satisfiability, which the system answers, so the system would answer it. Hence it is not in NP, which by definition means that its yes-instances have no certificate of polynomial size checkable in polynomial time. An undecidable question is outside NP a fortiori, since every question in NP is decided by trying every certificate of the bounded size; so again there is no polynomial verifier. □

Remark 5.6 (Scope of Proposition 5.5). (1) New questions can still be composed. Negation turns satisfiability into its complement, a question in co-NP, which falls under the proposition as a question outside NP unless NP and co-NP coincide; that is open. A proof that a particular question in co-NP has no polynomial verifier would separate NP from co-NP, and with them P from NP. *Standard.* (2) The premise is not established for the middle card. Its correspondence between traces and formulas is proved for every trace but for one step relation, exclusive or within a cell; realizing satisfiability in general requires it to encode the step of an arbitrary verifier. *Recorded*, as owed. (3) The proposition quantifies over every question and every verifier, and Proposition 2.3 bounds what a checker certifies about its own boundary; it is recorded here from cited results and is not among the middle card's receipts.

Proposed: a question with no polynomial verifier is one for which no record carries a receipt, and its answer, if supplied, is supplied as the quarter is, by a choice the record does not make.

6 The missing zero and the floor

The preceding sections locate the quarter by what records cannot do. This section formalizes the instances in which it appears as a term a record omits, and the instances in which it appears as a floor.

The origin of counting. The middle card defines thirty-six inductive records and a reader for each. Thirty-five readers count the base of their record as one cell: the origin is a shape, and it is counted. The reader of a measurement counts its origin as zero cells. *Measured*, extracted from the readers. This is the omission of Proposition 2.1: the $n = 0$ term that ψ leaves out and θ keeps, whose image in ζ is the residue 1 at $s = 1$, and which by Proposition 2.4(b) is the one value that restores the homogeneous transformation law. Euclid's assumption that a figure has one area however it is cut is the assumption that the origin carries something to count. *Proposed*: that the measurement's empty origin, the omitted term, and the Euclidean single shape are one choice of where counting starts.

The horizon. In a spatially flat expanding metric $ds^2 = -c^2dt^2 + a(t)^2d\mathbf{x}^2$ with Hubble rate $H = \dot{a}/a$, a photon moving radially toward the origin has comoving coordinate χ with $\dot{\chi} = -c/a$, and proper distance $D = a\chi$ with

$$\dot{D} = HD - c.$$

On the Hubble sphere $D = c/H$ the photon's proper distance is momentarily stationary: its approach velocity is exactly zero. If the Hubble sphere grows, the photon is later inside it, $\dot{D} < 0$, and it arrives. In pure de Sitter space, $a(t) = e^{Ht}$ with H constant, the Hubble sphere is fixed and coincides with the event horizon; $D = c/H$ is an equilibrium of $\dot{D} = HD - c$, and the photon never arrives [4]. *Standard.* The zero is a statement about proper distance: every comoving observer the photon passes measures its local speed as c, and a uniformly accelerated observer in flat space has a horizon of the same kind. *Standard. Proposed*: that the missing zero is that photon, counted one cell late where the sphere grows and never where it does not.

The multiplier. For a linear program $\min\{c^{\mathsf{T}}x : Ax \geq b,\ x \geq 0\}$ with dual multipliers $\lambda \geq 0$ on the constraints $Ax \geq b$, optimality gives complementary slackness,

$$\lambda_i \, (Ax - b)_i = 0 \qquad \text{for every } i :$$

a constraint met exactly may carry a nonzero multiplier, its shadow price, and a constraint with nonzero slack carries a zero multiplier. *Standard.* A penalty relaxation replaces the constraint by a finite penalty and enforces it only to a nonzero residual, the bracket the penalty leaves; the constraint is then not met exactly. In the unimodular presentation of gravity used by the numerical card, the cosmological constant is the multiplier of the constraint $\sqrt{-g} = 1$ that fixes the volume element [9]. *Standard.* The middle card's vocabulary names both cases: the constraint met exactly, and the bracket a penalty leaves. *Measured*, extracted from its declarations. *Proposed*: that the quarter is the shadow price of the one constraint the reading meets exactly.

Relative distance only. In an expanding universe only ratios $a(t_1)/a(t_2)$ of the scale factor are observed; setting today's value to one is a convention, the choice of one member of a family no measurement separates. *Standard.* The middle card compares pairs only as ratios, and the classes of

that comparison form a single point: every scale is identified with every other. Division then picks a canonical representative, by subtraction while the order holds, with no selection principle beyond the well-order of the naturals. *Measured*, extracted from the declarations, for the single point and for the absence of the choice axiom. *Proposed*: that the single point is the unobservability of the cosmic scale and the division's representative is the convention that sets it.

Two floors. The middle card keeps two counters: cells, its resolution in space, and heartbeats, the elaborator's count of its own work, its resolution in time. Their ratio is a Courant number, and below a noise floor taken on the machine a finer reading is noise. That floor depends on the machine. *Measured*, observed in runs. Two floors are independent of the machine.

The first is a Fourier floor: for a function and its Fourier transform the product of the widths is at least one half, and position and momentum are such a pair, $\Delta x\,\Delta p \geq \frac{1}{2}$ with $\hbar = 1$ [8]. *Standard.*

The second is an arithmetic floor. Let $r(y) = 13y^2 - 36y + 18$, the residual of the numerical card. Its coefficients have no common factor, and its discriminant $36^2 - 4 \cdot 13 \cdot 18 = 360$ lies strictly between 18^2 and 19^2, so it is not a square and r has no rational root. For integers p and $q \geq 1$,

$$q^2\, r(p/q) = 13p^2 - 36pq + 18q^2$$

is a nonzero integer, hence $|r(p/q)| \geq 1/q^2$, and no fraction p/q lies closer to a root of r than a constant over q^2 [7]. *Standard*; the two comparisons are *measured* also in the middle card, extracted from its declarations. This is the constructive form of the proof that $\sqrt{2}$ is irrational: not that no fraction equals the root, but that every fraction misses it by at least a stated amount. The numerical card certifies the complementary upper half, a bracket narrower than one printed place at the reading. The error of the reading is therefore bracketed from both sides: below one place, and never zero. *Proposed*: that the uncertainty principle is the machine-dependent floor made independent of the machine.

Affixing the quarter to the bent card. The bent card is the card the mark takes to be followable. Every card of the middle card has a slot for its quarter, a number, beside its symbol; filling the slot is a single definition, and no part of the card's mark reads the slot, so the same mark sits on the card whatever number is affixed. *Measured*, extracted from the card's definitions. The card also has an index, the kind of thing it is a card of, and no field reads the index either. Let ι be the carrier map that sends a card to the card whose index is that card. The middle card builds the image of its carrier of facts under ι, places its mark on it unchanged, and deals its thirty-eight steps through it. Every step exists on the image. The first twelve are identical to the steps dealt without it, and the identity bills no axiom. At the thirteenth, where one construction lifts its quantum one universe above the box, the box is now a card and the affixed card appears in the quantum, so that step and every later one is a different proof. *Measured*, by dealing all thirty-eight steps through the image and comparing each. The card between is the bent card with the bent card affixed: no step distinguishes them through the twelfth, and the lift at the thirteenth is where the affix appears.

The two counting layers determine what affixing changes. The bent card counts applications of one symbol, a tally: a value affixed to each step changes nothing the tally reads, because the tally reads only the number of steps. The middle card's second layer, which takes one clicking cell as its unit, counts the cells that click: a fact affixed to each step is exactly what the second layer reads. The two counts agree if and only if every affixed fact holds; on the middle card's own tapes they agree, because every cell written there clicks. *Recorded*, from the two counting functions; *measured* for the agreement on the tapes.

A quarter can therefore be affixed to the bent card freely, and the affix establishes nothing. The card carries what it is given; the tally does not distinguish it; a reader of clicks reads the affixed fact and not its provenance. What is fixed under the card is supplied, not read, which is Theorem 3.1 at the table: the value affixed is a choice. *Proposed*: that the second counting layer is the first with a quarter affixed at every step.

The reader that deducts the quarter. Let a tape t be folded onto a slot q, so that the record is the word $t \cdot q$. Define two readers of length,

$$L_{\text{exact}}(t \cdot q) = |t \cdot q| - |q| = |t|\,, \qquad L_{\text{keep}}(t \cdot q) = |t \cdot q| = |t| + |q|\,.$$

The first reads the tape's length whatever the slot holds. *Proved here.* Read through the card between by L_{exact}, the reading agrees with the middle card's to every printed place, and the check is the stored walk read again, not repeated, in under a second. *Measured*, observed in a run. With the slot one cell long, L_{keep} reads every length one cell long; the step it drives no longer divides by its derivative, its fixed point near the reading repels, and the run grows without halting until it is stopped. *Measured*, observed in a run, with the algebra in the bent card, §5. The quarter is harmless exactly when it is deducted.

One count, read three ways. The middle card's press adds one page at a time, and each page adds one of each: a cell to the tape it carries, a step to its counter, and a universe to the register it lifts. At its thirty-sixth page the counter reads thirty-six, the tape thirty-seven, and the register stands thirty-five universes up. *Measured*, by reducing the press. Page, length and universe height are one count, separated by a fencepost: one past, one short. With one symbol on every tape there is one tape of each length, so the length of a tape is the enumeration of the records, and counting and enumerating coincide. *Proposed*: that this is the enumeration map of the questions whose answers can be confirmed by a run.

Every quarter is one quarter. Every card the middle card deals holds the same quarter: one cell, the zero whose fact is $T = T$. The card of propositions, the card of facts, the card between, and the page on which the reduction is written all hold it. *Measured*, extracted from the definitions. The interaction of a card never reads its slot, and no card exists without one: the quarter is a witness that must exist and is never inspected, which is Theorem 3.1 in its smallest form. The press carries it to the top unchanged. Each page lifts it one universe and adds nothing, so at the thirty-sixth page it stands thirty-five universes up and is still the one cell; returned to the card between, it gives the card between. *Measured*, extracted from the pages.

Two quarters from the inference. The step that names inference carries two closures: a conclusion that holds a constructed reading apart from a read one, and a closure that ends on the agreement of the two tables. Read out as tapes, the first is one cell resting on the card's own mark and the second three cells resting on that agreement: the first quarters in the card that do not rest on $T = T$. Cards holding them read the same walk, so a measure does not distinguish them; their quarters differ in length, so a counter does. *Measured*, extracted from the declarations; the check bills no choice. *Proposed*: that this is the cut of the three-card monte, one deck to the measure and two to the count. The paradoxical decomposition itself, which would require choice, is not written.

The jar holds both. The middle card begins its bisection from two walls computed from its own lengths. In the units of the reading they are 129.6 and 137.7. Both the reading, 137.011 29 . . ., and the recommended value, 137.035 999 177(21) [14], lie between them. *Measured*, extracted from the declarations. At each midpoint the sign of the quadratic's residual determines which wall moves, so the quadratic selects between the two values. The separation is therefore not between the range of the construction and the laboratory's: it is between the laboratory value and the root of the polynomial, inside a range the two share.

Measured against tests. A Galerkin solution satisfies its equation exactly against every test function in its space, and a weak solution may differ from a strong one at every point while agreeing on every test. *Standard.* The middle card's measurement is of that kind. It does not read a value the machine computes; it counts the work of a whole run from outside, against a blank, and stops where a finer reading would be noise. Against that test the run is exact to the floor. *Measured*, observed in runs. The printed digits are the representative the division chose, and no measurement in the card tests them. Every agreement the card makes of itself closes by reflexivity, the two sides the same term, and bills nothing. *Measured*, extracted from the declarations. *Proposed*: that a laboratory measurement is a test of the same kind, so that the prediction is agreement and the number is its representative.

The difference. Agreement alone carries no information: the classes in which everything is identified form a single point, and that point accepts every number. *Measured*, extracted from the declarations. Information enters only where two things differ: a mark that clicks, a bracket a penalty leaves, the slack that slips on the second turn. The reading and the recommended value differ by 1.80 parts in ten thousand. Read as error, the quadratic is inexact and the measurement corrects it. Read as residue, the difference is the true-equals-false stanza of the argument against the world, the slip left after the second turn, and the quarter. Both readings are consistent with every receipt in the three cards, and no card chooses between them, since the choice would be the record deciding what it is about. The reading is the knower's choice. *Proposed.*

Order without counting. The natural numbers have two descriptions. One is a zero and a successor: every number is a term built in finitely many steps, so to have a number is to have built it. The other is a zero and a total order: a least element and a relation that ranks everything. The second states how numbers compare and says nothing about whether each can be written down; it presupposes the representation without counting it. No first-order description of that kind determines the naturals: every such theory with the naturals as a model also has models with elements beyond every finite one [18]. *Standard.* What excludes them is representation, a type whose only members are finitely built terms. The middle card keeps the two apart. Its orders are hypotheses, and before a number is used, divided, printed, or set against a measurement, it is exhibited as a tape with a reader that reads its length; the card does not use the proof assistant's own natural numbers until its tapes have produced them. *Measured*, extracted from the card's construction. The language can therefore hold the counterfactual that a number is not represented, and the card exhibits each representation before it relies on it. Defining the two together, as the proof assistant's own numbers do, is the bent card: existence handed in before it is counted.

The bug as a counterfactual. The same separation removes a question that would otherwise precede the reading: whether the checker is the most efficient possible, or carries a bug the device

only offsets. Neither answer is required. A bug can be stated by order alone, as something that ranks somewhere without being exhibited, which is a counterfactual the language can hold. It is then searched for in the finite space the checker's iterations visit. That space is bounded and its size is a count, not a time: the meter halts a run at a fixed multiple of its reference plus a measured floor, and the work of one forced symbol is the same number of the checker's units, four thousand two hundred and eighty-six, on two different machines. *Measured*, observed in runs. If the search finds the bug, the bug is a tape, and exists. If the bound is reached first, the counterfactual remains a counterfactual and is read as nothing, which is the loop considered harmless. In either case the efficiency of the checker does not enter: efficiency is a property of the clock, and the search is measured in the count. *Recorded*, from the meter's bound and the measured count. *Proposed*: that every bug the device offsets is one such counterfactual, found or not within the bound.

7 Where the quarter is in the three cards

Table 1 locates the quarter in each card: the coordinate in which it sits, the receipt the card retains for it, and the form of the truncation or projection through which the card carries it.

Card	Coordinate of the quarter	Retained receipt	Truncation or projection
Geometric	the fourth ambient coordinate w of $S^3 \subset \mathbb{R}^4$	the sheet sign $\sigma \in \{+1, -1\}$ of w, a receipt off the silhouette $\{w = 0\}$	$P : S^3 \to \overline{B}^3$, $P(x, w) = x$; fibers of size two over the open ball, one on the silhouette
Numerical	the root $\alpha = (18 + 3\sqrt{10})/13$ of r, read as $g_x\alpha = 972(6 + \sqrt{10})/65$ with $g_x = 324/5$; the constant Λ	the branch $y > 18/13$ and the rational bracket walls L_{108}, U_{108} with equal floors; for Λ, the contracted Bianchi identity and the datum that fixes it	the certified twenty-four-place truncation $\lfloor Sg_xL_{108} \rfloor = \lfloor Sg_xU_{108} \rfloor$, $S = 10^{24}$; for Λ, the term $\Lambda g_{\mu\nu}$
Middle	the name, and the slot beside each symbol, one cell whose fact is $T = T$	the deduction of the slot's length, $\lvert t \cdot q \rvert - \lvert q \rvert$; the ledger a check confirms line by line	the length of a tape whose every cell is the same symbol

Table 1: The quarter in the three cards. Geometric and numerical entries are *recorded* from those cards' propositions; middle-card entries are *measured*, extracted from its declarations.

The geometric card carries the whole in four coordinates and observes three. The fourth survives as the image inside the ball together with a sign that records the sheet; without the sign, two states have one image. Its quarter is the fourth coordinate, and its receipt is the sign.

The numerical card carries a certified reading to twenty-four places and a cosmological constant as a residue. Its quarter is the root: finitely representable as an algebraic number, never as a terminating decimal, so its digits are a countable series that does not close. The order on $\mathbb{R}$ selects the branch; the two rational walls of the bracket are the receipt; the twenty-four printed places are the projection, certified exactly to the resolution at which the bracket lies in one reporting cell and no further. The datum that fixes Λ is its second quarter, with the Bianchi identity as receipt. *Recorded* for the bracket and the truncation, from the numerical card's certificate. *Proposed*: the reading of the root as the quarter.

The middle card carries a tape whose every cell is the same symbol, and a census that reads only what has a name. Its quarter is the name: the compiler can require one and cannot mint one. *Measured*.

In all three the record is complete without the quarter and incomplete with respect to it, which is Proposition 2.4 in each card's coordinates.

The argument card, the patter, places the quarter twice more. Of the five posts that bound its four reals, four are written by a record and the fifth, the post at zero, is written by none; its value enters additively, so its fibre is all of the reals. And the table's antisymmetric part is the bent card times a quarter turn R, $R^2 = -1$, which read as i gives the compact turns, and on a box of four is also the time matrix of the Dirac algebra. *Measured* for R, its square and the time matrix; *proposed*, there, that the post at zero is this card's quarter, and not claimed that the quarter turn and the quarter are one.

This card has no press check, because a press check compares two readings of a register and the quarter is what the readings are of. It has no counter, because nothing is written here that was not already carried. It has no page one.

The last line. The number the three cards straddle agrees with a measured constant to four places. As far as the literature records, there is no accepted mathematical derivation of the fine-structure constant. *Standard*, as a report on the literature: the constant is measured [14], and no derivation of it is accepted. The resemblance is *proposed*, as it is throughout. This card adds the observation that a number with no derivation and a number with no record in a specified family of readers are, to a record of that family, the same kind of object: a projection, a receipt, and a choice.

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